Tarnwell LX-200

Field Service Handbook

Published by Larkfield Instruments Ltd, 14 Harrow Mill Lane, Larkfield, LF3 9QA.

This handbook explains how field engineers install, calibrate and maintain the Tarnwell LX-200 water quality monitor. It is written for trained service staff and for site operators who assist them.

1. Introduction

The Tarnwell LX-200 measures pH, conductivity, dissolved oxygen and turbidity in open channels and treatment tanks. Readings are logged every minute and sent to the site gateway over a wired or wireless link.

Each chapter starts with the tools and parts you need. Values in this handbook apply to firmware 4.2 and later. Earlier firmware uses the same procedures with fewer menu options.

2. Safety

Read this chapter before you open the enclosure. The monitor runs on 24 V DC, but the site cabinet that feeds it may carry mains voltage.

2.1 Personal protective equipment

Wear safety glasses, nitrile gloves and waterproof boots. Use a life jacket when you work within two metres of an open channel deeper than one metre.

2.2 Isolation procedure

Switch off the cabinet breaker, lock it and attach a personal tag. Check that the status light on the monitor is off before you disconnect any cable. Never rely on the monitor's own power button for isolation.

2.3 Chemical handling

Calibration buffers are mildly corrosive. Store them below 25 °C, keep the bottles closed and rinse any spill with clean water. Safety data sheets are kept in the van folder.

3. Installation

3.1 Site survey

Confirm with the site operator where the sensor head will be mounted. Choose a point with steady flow, at least 30 cm below the lowest expected water level and away from aeration outlets that trap bubbles on the optical window.

3.2 Mounting the sensor head

Fix the stainless steel bracket to the channel wall with the four M10 anchors supplied. Slide the sensor pole into the bracket and tighten the clamp until the pole cannot rotate by hand. Route the cable through the strain relief.

3.3 Connecting the gateway

Connect the data cable to port B of the gateway. The gateway detects the monitor within two minutes and shows it with its serial number, for example LX2-0457-1182.

4. Calibration

Calibrate every probe after installation, after any probe replacement and at the interval given in chapter 5. The monitor stores the last ten calibration records.

4.1 pH probe

Use a two-point calibration with pH 7.00 and pH 4.01 buffers. Rinse the probe with deionised water between buffers and wait until the reading is stable for 30 seconds.

4.2 Conductivity probe

Use the 1413 µS/cm standard. Hold the probe in the middle of the beaker, away from the walls, and gently tap it to release trapped air.

4.3 Turbidity sensor

Wipe the optical window with a lint-free cloth, then run the zero calibration in clean water. Accept the result only if the drift reported by the monitor is below 0.2 NTU.

5. Maintenance schedule

The table lists the routine tasks per interval. Record every visit in the service log of the gateway so that the next engineer sees what was done.

Interval	Task	Parts	Time
Monthly	Clean the optical window and check the wiper blade	Cleaning kit CK-01	15 min
Quarterly	Two-point pH calibration and conductivity check	Buffer set BS-07	40 min
Six-monthly	Replace the wiper blade and the desiccant pack	WB-12, DP-03	25 min
Yearly	Replace the pH probe junction and inspect all seals	Junction kit JK-20	60 min
Every 3 years	Replace the dissolved oxygen membrane cap	Membrane cap MC-05	30 min

Shorten the intervals by half on sites with heavy algae growth or with suspended solids above 200 mg/l.

6. Troubleshooting

6.1 No readings on the gateway

Check the status light. Steady green means the monitor runs and the fault is on the link. Flashing amber means the monitor waits for a calibration. No light means there is no power: check the cabinet breaker and the 24 V supply.

6.2 Unstable pH readings

Unstable readings usually come from a dirty or dry junction. Soak the probe in storage solution for one hour and calibrate again. Replace the junction if the slope stays below 92 percent.

6.3 Turbidity reads too high

Look for bubbles or a biofilm on the optical window. Check that the wiper parks outside the measuring path. Move the sensor if aeration outlets are close.

7. Service requests

Site operators request a visit with the service request form. The form is a fillable PDF: the operator types the details, signs it and sends it to the regional service desk.

7.1 Priorities

Standard: a visit within ten working days. **Urgent**: within two working days, for a monitor that still measures but drifts. **Emergency**: the same day, for a monitor that stopped measuring on a regulated discharge point.

7.2 Contact

Regional service desk, Larkfield Instruments Ltd, service@larkfield.example, telephone 01632 960 418 (fictitious number).

8. Glossary

Buffer. A solution with a known pH value used to calibrate a pH probe.

Drift. A slow change of a reading over time without a change of the measured water.

Gateway. The site unit that collects readings from monitors and sends them to the data platform.

Junction. The porous part of a pH probe that connects the reference electrode to the water.

NTU. Nephelometric turbidity unit, the unit used for turbidity readings.

Slope. The response of a pH probe compared with an ideal probe, in percent.

All names, addresses, part numbers and telephone numbers in this document are fictitious.